

Md. Irtiza Hossain

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in LinkedIn 🐙 GitHub

Research Interests

Computer Vision, Human Behaviour Analytics, Scene Understanding, Visual Reasoning with Graph Neural Networks, Multimodal Representation Learning, Uncertainty-Aware and Explainable AI, 3D Perception for Autonomous Systems, Surveillance Intelligence, and Human-Centric Artificial Intelligence.

Education

Master of Science in Computer Science and Engineering [Specialization in Data Science] *Jun 2025 – Present*
BRAC University, Dhaka, Bangladesh

Bachelor of Science in Computer Science and Engineering *Dec 2019 – May 2024*
BRAC University, Dhaka, Bangladesh

- **CGPA:** 3.90 / 4.00
- **Thesis:** *Enhanced CNN Approaches for Multi-Image Embedding in Image Steganography*
- **Honours:** Vice Chancellor's List (5 semesters); Dean's List (2 semesters)

Publications

Published

- **Edge-Native Digitization of Handwritten Marksheets: A Hybrid Heuristic-Deep Learning Framework** 2026
Md. Irtiza Hossain, Junaid Ahmed Sifat, Abir Chowdhury, Md. Imran Bin Azad
QPain 2026
[IEEE Xplore](#) — [Code](#)
- **Enhanced CNN Approaches for Multi-Image Embedding in Image Steganography** 2024
Md. Irtiza Hossain, Samiul Kadir, Farhan Ishraq Fagun, Ishtiaq Samiul, Rafi Zaman Saukhin
ICICT 2024
[IEEE Xplore](#) — [Code](#)

Under Review & In Preparation

- **Predictive Self-Conditioning for Accurate and Calibrated LiDAR 3D Object Detection**
Sami Sadat, Md. Irtiza Hossain, Shaownak Md. Ibne Shahriar Talukder, Md. Golam Rabiul Alam
Under Review (WACV 2027)
- **SmokeGraph-Bench: A Benchmark for Smoking Behaviour Detection with Geometric Relational Baselines**
Md. Irtiza Hossain, Sami Sadat, Junaid Ahmed Sifat, Md. Waseq Alauddin Alvi, Md. Khalilur Rahman
Under Review (EAAI) [Preprint](#)
- **CLEAR-MoE: Shared-Basis Expert Extraction from Frozen Vision Transformers via Calibration-Driven Layer Selection**
Md. Irtiza Hossain, Humaira Ayesha, Junaid Ahmed Sifat
Under Review (ICIT 2026) [arXiv](#) — [Code](#)
- **Uncertainty-Aware and Efficient 3D Object Detection for Autonomous Driving: A Comprehensive Survey**
Sami Sadat, Md. Irtiza Hossain, Md. Golam Rabiul Alam
In Preparation

Teaching Experience

Adjunct Lecturer

Nov 2024 – Present

BRAC University, Dhaka, Bangladesh

- Delivered lectures and supervised laboratory sessions for multiple undergraduate courses, including Data Structures, Algorithms, Artificial Intelligence, Machine Learning, Computer Interfacing, Database Management Systems, System Analysis & Design, and Introduction to Robotics.

Undergraduate Teaching Assistant

Oct 2022 – May 2024

BRAC University, Dhaka, Bangladesh

- Assisted instruction in *Programming Language II (Object-Oriented Programming)* (Feb 2023 – May 2024) and *Differential Calculus and Coordinate Geometry* (Oct – Dec 2022); held regular office hours and graded examinations.

Industry Experience

Junior Software Engineer

Aug 2024 – Nov 2024

Orbit Informatics, Dhaka, Bangladesh

- Architected and delivered a production Chrome extension with a React frontend and Node.js backend, enabling dynamic browser integration for enterprise users.

Trainee Engineer (.NET)

Jun 2024 – Aug 2024

PRAN-RFL Group (*othoba.com*), Dhaka, Bangladesh

- Automated the abandoned-cart detection pipeline, contributing to a measured 1–2% increase in recovered sales revenue through targeted follow-up campaigns.

Selected Projects

- Explainable AI for Brain Cancer Classification from MRI Scans**

[Code](#)

Applied Grad-CAM and SHAP attribution methods on CNN classifiers to localise tumour regions and improve model transparency for clinical review.

- Telecom Churn Prediction**

[Code](#)

Built an ensemble ML pipeline (XGBoost, Random Forest, SHAP explanations) to identify at-risk subscribers from usage and billing features.

- Power X Gym — Full-Stack Web Application (MERN Stack)**

[Live](#)

Designed and deployed a gym management platform with role-based authentication, class scheduling, and payment integration.

Technical Skills

Research & ML Frameworks: PyTorch, TensorFlow, Keras, Scikit-learn, OpenCV, Detectron2, MMDetection, Hugging Face Transformers

Programming Languages: Python (primary), C, C#, JavaScript, PHP

Tools & Platforms: Git, Docker, Linux/Bash, L^AT_EX, Jupyter, Weights & Biases, Google Colab / CUDA-enabled GPU clusters

Web & Backend: React, Node.js, Redux, MongoDB, MySQL, Firebase, .NET, PHP

References

Md. Khalilur Rhaman, PhD

Professor

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BRAC University, Dhaka, Bangladesh

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Dr. Muhammad Iqbal Hossain

Associate Professor

Dept. of Computer Science & Engineering

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